

WAP to add new row to existing dataframe.

Input :

```
jyotirmay adding a new row to existing dataframe.py - E:\11th codes\jyotirmay adding a new row to existing dataframe.py (3.13.14)
File Edit Format Run Options Window Help

import pandas as pd

data = [
    ["Alice", 25, "New York"],
    ["Bob", 30, "London"],
    ["Charlie", 35, "Paris"]
]

df = pd.DataFrame(data, columns=["Name", "Age", "City"])
print(df)
```

Output :

```
== RESTART: E:\11th codes\jyotirmay adding a new row to existing dataframe.py ==
      Name  Age  City
0   Alice   25  New York
1    Bob    30   London
2  Charlie   35    Paris
>>>
```

WAP to create a dataframe.

Input :

```
jyotirmay creating a dataframe.py - E:\11th codes\jyotirmay creating a dataframe.py (3.13.14)
File Edit Format Run Options Window Help

import pandas as pd

data = {
    "Name": ["Alice", "Bob", "Charlie"],
    "Age": [25, 30, 35],
    "City": ["New York", "London", "Paris"]
}

df = pd.DataFrame(data)
print(df)
```

Output :

```
===== RESTART: E:\11th codes\jyotirmay creating a dataframe.py
```

```
      Name  Age   City
0    Alice   25 New York
1     Bob   30   London
2  Charlie   35   Paris
```

WAP on attributes of dataframe.

Input :

```
jyotirmay attributes of dataframe.py - E:\11th codes\jyotirmay attributes of dataframe.py (3.13.14)
File Edit Format Run Options Window Help

import pandas as pd

data = {
    'Name': ['Ankit', 'Rahul', 'Karan'],
    'Age': [18, 19, 17],
    'Salary': [5000, 6000, 4500]
}

df = pd.DataFrame(data)

print("Index:", df.index)
print("Columns:", df.columns)
print("Axes:", df.axes)
print("Data Types:", df.dtypes)
print("Size:", df.size)
print("Shape is:", df.shape)
print("Empty:", df.empty)
print("Heading 2 rows are:")
print(df.head(n=2))
print("Last 2 rows are:")
print(df.tail(n=2))
```

Output :

```
===== RESTART: E:\11th codes\jyotirmay attributes of dataframe.py =====
Index: RangeIndex(start=0, stop=3, step=1)
Columns: Index(['Name', 'Age', 'Salary'], dtype='str')
Axes: [RangeIndex(start=0, stop=3, step=1), Index(['Name', 'Age', 'Salary'], dtype='str')]
Data Types: Name      str
Age      int64
Salary   int64
dtype: object
Size: 9
Shape is: (3, 3)
Empty: False
Heading 2 rows are:
   Name  Age  Salary
0  Ankit   18   5000
1  Rahul   19   6000
Last 2 rows are:
   Name  Age  Salary
1  Rahul   19   6000
2  Karan   17   4500
```

WAP to delete rows from existing dataframe.

Input :

```
jyotirmay deleting rows from existing dataframes.py - E:\11th codes\jyotirmay deleting rows from existing dataframes.py (3.13.14)
File Edit Format Run Options Window Help
import pandas as pd

data = {
    'Name': ['Amit', 'Bhaum', 'Chetan'],
    'Marks': [90, 85, 92],
    'City': ['Delhi', 'Mumbai', 'Delhi']
}

df = pd.DataFrame(data, index=['R1', 'R2', 'R3'])

print("Original DataFrame:")
print(df)

print("\n")

df = df.drop('R2')

print("DataFrame after deleting R2:")
print(df)
```

Output :

```
== RESTART: E:\11th codes\jyotirmay deleting rows from existing dataframes.py ==
Original DataFrame:
   Name  Marks  City
R1  Amit     90  Delhi
R2  Bhaum     85  Mumbai
R3  Chetan     92  Delhi

DataFrame after deleting R2:
   Name  Marks  City
R1  Amit     90  Delhi
R3  Chetan     92  Delhi
```

WAP program to find the mean of the following grouped frequency distribution using the mid-point method.

Input :

File Edit Format Run Options Window Help

```
mid = [15, 35, 55, 75, 95]
fr = [1, 2, 3, 4, 5]

fx = [f * m for f, m in zip(fr, mid)]

print("After zipping:", fx)

sum_fx = sum(fx)
sum_fr = sum(fr)

mean = sum_fx / sum_fr

print("Mid points:", mid)
print("Frequencies:", fr)
print("Sum of frequency * mid points:", sum_fx)
print("Sum of frequency:", sum_fr)
print("Mean of grouped data:", mean)
```

Output :

```
===== RESTART: E:\11th codes\jyotirmay frequency distribution in grouped data.py
After zipping: [15, 70, 165, 300, 475]
Mid points: [15, 35, 55, 75, 95]
Frequencies: [1, 2, 3, 4, 5]
Sum of frequency * mid points: 1025
Sum of frequency: 15
Mean of grouped data: 68.33333333333333
```

WAP using an if-else-elif chain to display the appropriate meal menu based on the meal time.

Input :

```
jyotirmay if else chain.py - E:\11th codes\jyotirmay if else chain.py (3.13.14)
File Edit Format Run Options Window Help
meal_time = 14
if meal_time <11:
    print("serving breakfast menu")
    menu_type = "breakfast"
elif meal_time <16:
    print("serving lunch menu")
    menu_type = "lunch"
elif meal_time < 22:
    print("serving dinner menu")
    menu_type = "dinner"
else:
    print("sorry , kitchen is closed")
    menu_type = "closed"
```

Output :

```
===== RESTART: E:\11th codes\jyotirmay if else chain.py
serving lunch menu
|
```

WAP on list manipulation.

Input :

 jyotirmay list manipulation.py - E:\11th codes\jyotirmay list manipulation.py (3.13.14)

File Edit Format Run Options Window Help

```
l1 = [5, 6, 7]
l2 = [1, 2, 3]
print(l1+l2)
l1.append(56)
print(l1)
l2.extend(l1)
print(l2)
print(l1*3)
|
```

Output :

```
===== RESTART: E:\11th codes\jyotirmay list manipulation.py
[5, 6, 7, 1, 2, 3]
[5, 6, 7, 56]
[1, 2, 3, 5, 6, 7, 56]
[5, 6, 7, 56, 5, 6, 7, 56, 5, 6, 7, 56]
|
```

WAP to demonstrate the use of different mathematical functions of the math module.

Input :

```
jyotirmay math python library.py - E:\11th codes\jyotirmay math python library.py (3.13.14)
File Edit Format Run Options Window Help
import math
print(math.pow(3,4))
print(math.sqrt(4))
print(math.ceil(45))
print(math.floor(45))
|
```

Output :

```
===== RESTART: E:\11th codes\jyotirmay math python library.py
81.0
2.0
45
45
|
```

WAP to find the highest number in a list.

Input :

```
jyotirmay maximum number finder.py - E:\11th codes\jyotirmay maximum number finder.py (3.13.14)
File Edit Format Run Options Window Help
l = [12,28,75,2,10]
max = 0
for i in l:
    if max < i:
        max = i
print("largest value", max)
|
```


Output :

```
===== RESTART: E:\11th codes\jyotirmay maximum number finder.py =====  
largest value 75
```

WAP to run nested loop.

Input :

 jyotirmay nested loop.py - E:\11th codes\jyotirmay nested loop.py (3.13.14)

File Edit Format Run Options Window Help

```
for i in range(1, 10):  
    if i == 5:  
        print("Outer loop i is 5")  
  
    for j in range(1, 10):  
        if j == 5:  
            continue  
        print("Inner j:", j)
```

Output :

```
===== RESTART: E:\11th codes\jyotirmay nested loop.py :
Inner j: 1
Inner j: 2
Inner j: 3
Inner j: 4
Inner j: 6
Inner j: 7
Inner j: 8
Inner j: 9
Inner j: 1
Inner j: 2
Inner j: 3
Inner j: 4
Inner j: 6
Inner j: 7
Inner j: 8
Inner j: 9
Inner j: 1
Inner j: 2
Inner j: 3
Inner j: 4
Inner j: 6
Inner j: 7
Inner j: 8
Inner j: 9
```

WAP on python list slicing.

Input :

```
jyotirmay python list slicing.py - E:\11th codes\jyotirmay python list slicing.py (3.13.14)
File Edit Format Run Options Window Help
l = [12,13,14,15,16,17]
print(l)
print(l[1:5])
print(l[::-1])
```

Output :

```
===== RESTART: E:\11th codes\jyotirmay python list slicing.py
[12, 13, 14, 15, 16, 17]
[13, 14, 15, 16]
[17, 16, 15, 14, 13, 12]
```

WAP to demonstrate the use of NumPy arrays and generate a random NumPy array.

Input :

```
jyotirmay python list.py - E:\11th codes\jyotirmay python list.py (3.13.14)
File Edit Format Run Options Window Help
import numpy as np
import random
my_list = [1, "puja", 34, 67, 78]
print(my_list)
ar = np.array([1, 2, 3, 4, 5])
print(ar)
a = np.random.randint(1, 10, size=(3, 4))
print(a)
```

Output :

```
===== RESTART: E:\11th codes\jyotirmay python list.py  
[1, 'puja', 34, 67, 78]  
[1 2 3 4 5]  
[[4 4 1 5]  
 [3 4 9 1]  
 [4 8 9 3]]
```

WAP to demonstrate the use of different functions of the random module.

Input :

```
jyotirmay random python library.py - E:\11th codes\jyotirmay random python library.py (3.13.14)
File Edit Format Run Options Window Help
import random
print(random.random())
print(random.randrange(2, 5))
print(random.randint(2, 5))
```

Output :

```
===== RESTART: E:\11th codes\jyotirmay random python library.py :
0.043339073876541456
3
2
```

WAP to find the smallest number in an array.

Input :

```
jyotirmay smallest number in array.py - E:\11th codes\jyotirmay smallest number in array.py (3.13.14)
File Edit Format Run Options Window Help
import numpy as np
arr = np.array([250, 300, 350, 400, 450])
small = arr[0]
for i in arr:
    if i < small:
        small = i
print("Smallest value:", small)
```

Output :

```
===== RESTART: E:\11th codes\jyotirmay smallest number in array.py
Smallest value: 250
|
```

WAP to demonstrate the use of different functions of the statistics module.

Input :

```
jyotirmay statistics python library.py - E:\11th codes\jyotirmay statistics python library.py (3.13.14)
File Edit Format Run Options Window Help
import statistics
l = [1,34,56,78,90,12,67]
print(statistics.mean(l))
print(statistics.mode(l))
print(statistics.median(l))
```

Output :

```
===== RESTART: E:\11th codes\jyotirmay statistics python library.py
48.285714285714285
1
56
```

WAP to demonstrate the use of different string functions and operations.

Input :

```
jyotirmay string manipulation program.py - E:\11th codes\jyotirmay string manipulation program.py (3.13.14)
File Edit Format Run Options Window Help
text="hi"
upper_text = text.upper()
lower_text = text.lower()
split_text = text.split()

print("original text:" , text)
print("split text:" , split_text)
print("length:", len(text))
print("first word:" , text [:6])
```

Output :

```
===== RESTART: E:\11th codes\jyotirmay string manipulation program.py =====
original text: hi
split text: ['hi']
length: 2
first word: hi
```

WAP to find the sum of all numbers in an array.

Input :

```
jyotirmay to find sum of all number in array.py - E:\11th codes\jyotirmay to find sum of all number in array.py (3.13.14)
File Edit Format Run Options Window Help
import numpy
import numpy as np
ar = np.array([250,300,350,400,450,500])
sum = 0
for i in ar:
    sum=sum+i
    print("total sales is" ,sum)
    av = sum/7
    print("average is" , av)
```

Output :

```
==== RESTART: E:\11th codes\jyotirmay to find sum of all number in array.py
total sales is 250
average is 35.714285714285715
total sales is 550
average is 78.57142857142857
total sales is 900
average is 128.57142857142858
total sales is 1300
average is 185.71428571428572
total sales is 1750
average is 250.0
total sales is 2250
average is 321.42857142857144
```